

FARHAN SHAIK

Data Science | Data Analytics | Data Engineering

New York, NY | farhanshaik422@gmail.com | +1 475-300-3393

ACADEMICS

University Of New Haven, Connecticut.
M.S. in Data Science, **CGPA** –3.6/4.0

Jan 2022 - May 2023

Osmania University, Hyderabad, India
B.E in Civil Engineering, **CGPA** – 7.9/10

Jul 2016 - Sep 2020

TECHNICAL SKILLS

Programming Languages: SQL, Python (NumPy, Pandas, scikit-learn, Matplotlib, TensorFlow, Pytorch), R, Pyspark

Statistics/Machine Learning: Regression, Classification, A/B Testing, Forecasting, Neural Networks, Time Series.

Natural Language Processing: Text Analysis, Name Entity recognition, Transformers, RNN, LSTM,

Data Analysis & Visualization Tools: Power BI, Tableau, Looker Studio, SAS, MS-Excel

Databases: MySQL, PostgreSQL, Microsoft SQL Server.

Version Control: Git, SVN

Cloud Technologies: AWS, Azure, GCP, Databricks, Snowflake.

PROFESSIONAL EXPERIENCE

Data Engineer – Hedge Funds– NY – USA

July 2024 – Present

- Work closely with **BI** and application teams to understand new data requirements and translate them into scalable architectural and technical specifications for ingesting, transforming, and delivering trusted data into the enterprise data ecosystem.
- Use **Azure Data Factory (ADF)** to automate and orchestrate end-to-end data pipelines by extracting data from multiple internal and third-party sources and loading it into **Azure SQL Server** Databases, ensuring secure, scalable, and reliable data movement across the enterprise.
- Build and maintain pipelines that ingest data from several key sources including Azure Blob Storage, Snowflake shared databases, **HubSpot API**, **Salesforce**, **SFTP** locations, and SharePoint via Power Automate workflows that move files into Azure Blob Storage.
- Performing data cleaning using **SQL**, **Python**, and data wrangling libraries like **Pandas**, preparing datasets for analysis by handling missing values, correcting inconsistencies, and transforming data into usable formats.
- Developing and optimizing complex **SQL** queries and scripts, using SQL Server to ensure efficient data retrieval and processing, minimizing runtime and improving performance.
- Develop exact matching processes to identify and link records from multiple data sources to existing Salesforce Contact objects, ensuring clean and consolidated data for reporting and analytics.
- Load daily updated **Salesforce** objects into Azure SQL Server, which serve as the primary source for the matching process.
- Implement exact matching logic on key columns such as email, and unique identifiers, creating mapping tables and stored procedures in SQL to standardize matches.
- Integrate exact matching stored procedures into ADF pipelines, ensuring matching is applied before data is loaded into Azure SQL Server.
- Design multi-layer Snowflake data architecture (**Staging /Modeled/Reporting**) to ensure consistent data lineage, governance, and optimized data flow across cloud analytics platforms.
- Develop **Snowflake** data models using dimensional modeling, star/snowflake schemas, and domain-driven design to support scalable analytics, data science workloads, and enterprise reporting.
- Utilize Liquibase CI/CD pipelines to deploy new or modified tables, views, functions, and stored procedures from **DEV to UAT and Production** environments, ensuring consistency, version control, and auditability across environments.

Data Analyst – NYSIF – NY – USA

Jan 2024 – July 2024

- Enhanced the performance of the high-cost insurance claims prediction model using **XGBoost**, improving **precision** by **22%**.
- Implemented **data collection, cleaning, and preprocessing** tasks to prepare large-scale insurance claims data for analysis, leveraging **Numpy** and **Pandas** to efficiently handle complex data analysis process.
- Collaborated with **actuaries** and **underwriters** to define business objectives, **key performance indicators (KPIs)**, and success criteria for the project.
- Developed and optimized **SQL** queries, **views**, and **functions** to meet business intelligence and reporting needs
- Conducted exploratory data analysis (**EDA**) to uncover patterns, **trends**, and **outliers** in the insurance claims data, providing insights to inform **model development** and **feature selection**.
- Performed statistical analysis to understand claim distribution, calculated **descriptive statistics**, explored **confidence intervals**, assessed **feature correlations**, and identified potential predictors of **high-cost claims**.
- Created insightful visualizations and interactive dashboards using **Spotfire** to communicate insights and model predictions to business stakeholders effectively.
- Implemented **feature engineering** techniques to create new variables and transform existing ones, improving the predictive performance of the models.
- Evaluated model performance using metrics such as **accuracy, precision, recall, F1-score, and ROC AUC**, iteratively refining models to achieve desired outcomes. Additionally, enhanced complex machine learning models with hyper-parameter tuning using cross-validation techniques like **Grid Search** and **K-Fold**.
- Implemented and maintained version control systems like **Git** to ensure efficient code management and collaboration.

Data Analyst – Emonics – NJ – USA

Aug 2023 – Jan 2024

- Created project performance reports and visualization on marketing data to address business requirements.
- Performed **ETL** and blending/wrangling on variety of data format from multiple disparate sources using complex **SQL** queries, **R** and **Azure Databricks** to enable stakeholders to gain business insights from data.
- Used **R** programming for data analysis and implementing forecasting models on large volumes of machine performance data to track customer power consumption pattern and used data visualization tools to generate report (**Power BI**).
- Facilitated and led workshops with cross-functional teams to define and prioritize data- driven projects.
- Designed and Developed **R-Shiny** app for fault detection in customers machine and generate condition reports.
- Analyzed complex data and identified anomalies and trends to provide useful insights.
- Analyzed existing **SQL** queries for improvements and developed custom **SQL** scripts to improve the Database efficiency, reduce data load time and enhance performance.
- Work on projects pertaining to in-depth analysis – trend analysis, historical data analysis and predictive analysis.

Data Analyst – Forge Technologies – NJ – USA

Feb 2023 – May 2023

- Conducted extensive data preprocessing tasks including handling **missing values, outliers**, and discrepancies to ensure data quality and consistency.
- Engineered and selected features relevant to the predictive model, improving the model's performance by **15%**.
- Performed **univariate and multivariate analysis on the data** to identify any **underlying pattern** in the data and associations between the variables.
- Built the machine learning model include: **SVM, random forest, XGboost** to score and identify the potential new business case with Python Scikit-learn.
- Applied regularization techniques to prevent **overfitting** of the model and optimizing the model
- Designed and Implemented **SQL Server** databases to store and retrieve application data for real time analysis
- Handled **stored procedures** to fix runtime bugs.
- Participated in code reviews and contributed to continuous improvement of coding standards and development practices.

Data Science Intern – Innodatatics – India

Jun 2021 – Nov 2021

- Actively involved in all phases of the data science project life cycle, such as **Data Collection, Data Cleaning, Data Visualization** and **Model development, Model Validation** and **Deployment** to deliver the data science solutions.
- Performed data cleaning including transforming variables and dealing with missing value and ensured data quality, consistency, integrity using **Pandas, NumPy** and **Matplotlib** for Data Visualization.
- Created custom dashboards and visualizations in **Power BI**.
- Implemented Machine Learning models in **Python** and **R**.

- Built **recommendation system on machine learning** to improve the sales and to reduce the customer churn.
- Worked on Customer Segmentation that involves dividing customers into distinct groups based on similar characteristics.
- Worked with **NLTK library** to NLP data processing and finding the patterns.
- Used **F-Score, Precision, Recall, and A/B testing** to evaluate Model **performance**.

PROJECTS

PDF Conversational Application: Engage and Interact with Multiple Documents

- Developed a personalized project utilizing Python libraries such as **Streamlit, PyPDF2, and langchain** to facilitate efficient chat interactions across multiple PDF documents, demonstrating proficiency in text processing and natural language understanding techniques.
- Implemented advanced features leveraging **Google Generative AI Embeddings and FAISS vector stores** to enhance chat functionality, ensuring accurate and efficient responses to user questions within the PDF documents.

InvoiceGen: Intelligent Invoice Information Extraction and Question Answering

- Developed a comprehensive invoice extraction and question-answering project using Python libraries like **Streamlit, Pillow, and Google Generative AI**.
- Implemented advanced functionality where users can ask questions about the uploaded invoice, leveraging Streamlit and Pillow for seamless interaction and image processing, thereby automating the extraction of key information from invoices and provides the accurate answers to user queries.

Image Segmentation:

- Collected sample images consisting of 3 classes cars, buildings and persons from the university and annotated them using CVAT tool and built a scratch **Segnet** a deep convolution **encoder decoder** architecture using **Pytorch** for training.
- The encoder extracts the features from the input image and decoder generates a segmentation map by up sampling the encoded features.
- Intersection over Union (**IoU**) metric used to evaluate the performance of segmentation model.

CERTIFICATIONS

- MLOps Essentials: Model Development and Integration, LinkedIn Learning
- MLOps Essentials: Model Deployment and Monitoring, LinkedIn Learning
- Generative AI For Beginners - Google Gemini & Google Cloud, Udemy
- Master Big Data Course, Udemy
- Machine Learning with Python, IBM
- Certified in Microsoft Azure Data Engineer Associate (DP-203).